

23M413	DISCRETE MATHEMATICS AND AUTOMATA THEORY	L	T	P	C
		3	1	0	4

MODULE I MATHEMATICAL LOGIC

9

Connectives – Statement Formulas and Truth tables- Well formed formulas- Tautologies
Equivalence of Formulas-Duality Law- Tautological Implications- Functionally complete sets
of connectives- other Connectives- Disjunctive and Conjunctive normal forms- Principals
of Disjunctive and Conjunctive normal Forms.

MODULE II INFERENCE THEORY AND PREDICATE CALCULUS

9

Rules of inferences – Consistency of premises and indirect method of proof – Predicate
Calculus- Variables and Quantifiers- Theory of Inference for the Predicate Calculus – Formulas
Involving more than one Quantifier.

MODULE III ALGEBRAIC STRUCTURES I

9

Rings, Integral domains and Fields - Properties - Polynomial Rings - Construction of Finite
Fields ; Irreducible Polynomials - Primitive element of a Finite Field - Primitive Irreducible
Polynomials.

MODULE IV ALGEBRAIC STRUCTURES II

9

Residue arithmetic for Computers - Coding theory - Error Detection - Correction –Hamming
Distance - Minimum distance - Group Code, Linear Code and Cyclic Codes–Single error
correction; Encoding and Decoding Techniques.

MODULE V FINITE AUTOMATA AND TURING MACHINES

9

Chomsky Classification of Grammars- Finite State Automata – Definition – Transition diagrams
– Deterministic & Nondeterministic Finite Automata – Finite Automata with ϵ moves Turing
Machines: Definition– Addition of unary numbers – Multiplication of unary numbers –
Recognition of words of the form $0^n, 1^n$.

COURSE COUTCOMES

At the end of the course, students will be able to

CO1: Apply logical reasoning in verifying the correctness and validity of simple instances of
valid logical arguments.

CO2: Learn the basic concepts of rings, integral domains and fields to acquire knowledge to
solve the problems.

CO3: Illustrate various algebraic structures and their properties and design encoding and
decoding procedures for error detection and correction.

CO4: To acquire knowledge in constructing system models which are the natural extension of
Automata that is used to devise decision procedures.

CO5: Learn the basic concepts of Inference theory, rings, integral domains and fields to solve
the problems.

CO6: Apply the properties of Algebraic structure and Turing machine techniques in solving
engineering problems.

TOTAL : 45 PERIODS+ 15 = 60 PERIODS

TEXTBOOKS

1. Grewal B.S., "Higher Engineering Mathematics", 44th Edition, Khanna Publishers - (2021).
2. Erwin Kreyszig, "Advanced Engineering Mathematics, 10th Edition (2015), John Wiley & Sons.

REFERENCE BOOKS

1. Joel R. Hass, Christopher E. Heil, Maurice D. Weir, "Thomas Calculus", 15th Edition (2022), Pearson.
2. B.V. Ramana, "Higher Engineering Mathematics", Kindle Edition (2018), Tata McGraw-Hill Publishing Company Limited, New Delhi.
3. Bali N, Goyal M. and Watkins C., "Advanced Engineering Mathematics", Firewall Media (An imprint of Lakshmi Publications Pvt., Ltd.), New Delhi, 7th Edition, 2015.

CO-PO & PSO MAPPING

CO	PO												PSO		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
1	3	3	1									1		1	
2	3	3	1									1		1	
3	3	3	1									1		1	
4	3	3	1									1		1	
5	3	3	1									1		1	
6	3	3	1									1		1	
Avg.	3	3	1									1		1	

1-low, 2-medium, 3-high

23AD411	MACHINE LEARNING TECHNIQUES	L	T	P	C
		3	0	0	3

MODULE I INTRODUCTION TO MACHINE LEARNING

9

Introduction- Types of Machine Learning- Application of Machine Learning- Hypothesis Space -Inductive Bias- Evaluation and Cross Validation.

MODULE II SUPERVISED LEARNING ALGORITHMS

9

Linear Models for Regression – Linear Models for Classification- Discriminant Functions, Probabilistic Generative Models, Probabilistic Discriminative Models – Decision Tree Learning – Bayesian Learning, Naïve Bayes –Ensemble Methods, Bagging, Boosting.

MODULE III UNSUPERVISED LEARNING ALGORITHMS

9

Clustering- Hierarchical Agglomerative Clustering-K-means – EM Algorithm- Mixtures of Gaussians –Dimensionality Reduction, Linear Discriminant Analysis, Factor Analysis, Principal Components Analysis, Independent Components Analysis.

MODULE IV NEURAL NETWORK AND BAYESIAN CONCEPTS

9

Introduction to Neural Network- Biological Neurons- Architectural of Neural Network- Implementation of ANN -Back propagation Algorithm – Importance of Bayesian Methods: Bayes Theorem - Bayes Theorem and Concept -Learning- Bayes Belief Network.

MODULE V CLUSTERING AND COMPUTATION LEARNING

9

Introduction to Clustering, Hierarchical Clustering –Agglomerative Clustering. Introduction to Computation Learning -Sample complexity: Finite Hypothesis Space- VC Dimensions- Introduction to Ensemble Learning.

COURSE COUTCOMES

At the end of the course, students will be able to

CO1: To understand the basic principles of machine learning techniques

CO2: To evaluate the strengths and weaknesses of machine learning algorithms

CO3: To Study the concepts of dimensionality reduction

CO4: To Summarize the concepts in Bayesian analysis from probability model and methods

CO5: To Design and implement neural network clustering algorithms and ensemble learning techniques

TOTAL : 45 PERIODS

TEXT BOOKS

1. I. A. Dhotre , Introduction to Machine learning, Technical Publication, 2022.
2. T. Mitchell, Machine Learning, McGraw Hill, 1997.

REFERENCE BOOKS

1. I. Goodfellow, Y. Bengio and A. Courville. Deep Learning. MITPress, 2016.

2. Richard S. Sutton and Andrew G. Barto, Reinforcement Learning: An Introduction, MIT Press, 2018.
3. M. P. Deisenroth, A. A. Faisal, C. S. Ong, Mathematics for Machine Learning, Cambridge University Press (1st edition), 2020.
4. S. Haykin. Neural networks and learning machines. Pearson 2008.
5. Christopher Bishop, Pattern Recognition and Machine Learning, Springer, 2006.
6. K. Murphy. Machine Learning: A probabilistic perspective, MIT Press, 2012.
7. Hastie, Tibshirani, Friedman, Elements of statistical learning, Springer, 2011.

CO-PO & PSO MAPPING

CO	PO												PSO		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
1	3	2		2									2		1
2		3		2											1
3			3	2	1										1
4			3	2	1										2
5			2	1									3	1	1
AVg.	3	2.5	2.6	1.8	1								2.5	1	1.2

1-low, 2-medium, 3-high

23AD412	WEB DEVELOPMENT TECHNOLOGIES	L	T	P	C
		3	0	0	3

MODULE I WEB SITE BASICS AND HTML

9

Web Essentials: Clients, Servers, and Communication. The Internet-Basic Internet Protocols -The World Wide Web-HTTP request message-response message-Web Clients Web Servers. Markup Languages: XHTML. An Introduction to HTML History-Versions-Basic XHTML Syntax and Semantics. Some Fundamental HTML Elements-Relative URLs-Lists-tables-Frames,Forms-HTML.

MODULE II CLIENT SIDE SCRIPTING

9

Style Sheets: CSS-Introduction to Cascading Style Sheets-Features-Core Syntax-Style Sheets and HTML- Style Rule Cascading and Inheritance-Text Properties-Box Model-Normal Flow Box Layout-Beyond the Normal Flow-CSS3.0. Client Side scripting - Introduction to Javascript – Arrays – Control statements –Events.

MODULE III SERVER SIDE SCRIPTING

9

Clustering- Host Objects: Browsers and the DOM-Introduction to the Document Object Model DOMHistory and Levels-Intrinsic Event Handling-Modifying Element Style-The DocumentTree-DOM Event Handling-Server-Side Programming: Java Servlets- Architecture -Overview-A Servlet-Generating Dynamic Content-Life Cycle- Parameter Data-Sessions-Cookies-URL Rewriting.

MODULE IV XML

9

Representing Web Data: XML-Documents and Vocabularies-Versions and Declaration-Namespaces- DOM based XML processing Event-oriented Parsing: SAX-Transforming XML Documents-Selecting XML Data: XPATH-Template based Transformations: XSLT-Displaying XML Documents in Browsers.

MODULE V JSP AND AJAX

9

Separating Programming and Presentation: JSP Technology Introduction-JSP and Servlets-Running JSP Applications Basic JSP-JavaBeans Classes and JSP-Tag Libraries and Files-Support for the Model-View-Controller Paradigm- Databases and JSP.AJAX: Ajax Client Server Architecture-XML Http Request Object-Call Back Methods-SOAP related Technologies.

COURSE COUTCOMES

At the end of the course, students will be able to

CO1: Design simple web pages using markup languages like HTML and XHTML.

CO2: Create dynamic web pages using DHTML and java script that is easy to navigate and use.

CO3: Program server-side web pages that have to process request from client-side web pages.

CO4: Represent web data using XML and develop web pages using JSP.

CO5: Understand the concept of application development using NODE JS

TOTAL : 45 PERIODS

TEXT BOOKS

1. Deital & Deital, "Internet and World Wide Web-How to Program", Pearson Education Fifth Edition, 2011.
2. Rashim Mogha, V.V Preetham. V.V. "Java Web Services Programming", Wiley Dreamtech, New Delhi, Java 2 Enterprise Edition, 2011. (Server Side Technologies)
3. Robert W.Sebesta, "Programming with World Wide Web", Pearson Education, Eighth Edition, 2015.(Web Based Applications and its Technologies)

REFERENCE BOOKS

1. Scot Johnson, Keith Ballinger,Davis Howard Chapman, "Special Edition Using Active Server Pages", Prentice Hall of India ,paperback 1999.
2. Ravi Kalakota and Andrew B Whinston, "Frontiers of e-commerce", Addison Wesley, 1st Edition, 2002.
3. Jeffrey C. Jackson," Web Technologies: A Computer Science Perspective", Pearson Education, Second Edition 2007

CO-PO & PSO MAPPING

CO	PO												PSO		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
1	3	3	3	2	3	1			2	2	3	2	3	3	3
2	3	3	3	3	3	1			2	2	3	2	3	3	3
3	3	3	3	3	3	1			2	2	3	2	3	3	3
4	3	3	3	3	3	1			2	2	3	2	3	3	3
5	2	2	2	3	3	1			2	2	2	2	3	3	3
AVg.	2.8	2.8	2.8	2.8	3	1			2	2	2.8	2	3	3	3

1-low, 2-medium,3-high

23AD413	DATA COMMUNICATION AND NETWORKING	L	T	P	C
		3	0	0	3

MODULE I PHYSICAL LAYER

9

Data Communications - Physical topology – OSI Model – TCP/IP Model – Bandwidth Utilization: FDM -TDM- Switching Techniques -Protocols and Standards- Transmission Media –Network components: Hubs - Switches - Routers - Gateways.

MODULE II DATA LINK LAYER

9

Data Link Layer - Design Issues - MAC addressing, MAC Protocols: Ethernet - Token ring, Error detection and correction techniques: Checksum - Cyclic Redundancy Check -Hamming code.

MODULE III NETWORK LAYER

9

IPv4 Addresses - Classful and Classless Addressing – Subnetting - Network Layer Protocols- IP – Unicast Routing –Routing Algorithms –Distance Vector Routing – Link State Routing – IPv6 Addressing - IPv6 Protocol – Transition from IPv4 to IPv6.

MODULE IV TRANSPORT LAYER

9

Introduction – Port and Socket Address - Transport Layer Protocols- Stop-and-Wait Protocol- Go-Back-N Protocol - Selective-Repeat Protocol - Bidirectional Protocols: Piggybacking – UDP –User Datagram - UDP Services - UDP Applications - TCP -Flow Control – Error Control - TCP Congestion Control.

MODULE V APPLICATION LAYER AND WIRELESS TECHNOLOGIES

9

World Wide Web and HTTP – FTP – Electronic Mail – Telnet -Domain Name System- Wireless Technologies: Wireless LAN- IEEE802.11- Wireless WAN: GSM, LTE- 4G/5G technologies.

COURSE COUTCOMES

At the end of the course, students will be able to

CO1:To evaluate the basics of the media, computer networks, and its layered architecture.

CO2:To summarize the error control, flow control mechanism and the medium access control.

CO3:To evaluate the working of the network layer, addressing, framing and routing of packets.

CO4:To analyse the principles of Transport layer and its protocols.

CO5:To understand the working of Application layer protocols and understand the working of Wireless Technologies

TOTAL : 45 PERIODS

TEXT BOOKS

1. Behrouz A Forouzan, "Data Communication & Networks", McGraw Hill, 5th Edition, 2013.
2. William Stallings, "Data & Computer Communications", Prentice Hall, 8th Edition, 2008.

REFERENCE BOOKS

1. Iti Saha Misra, "Wireless Communications and Networks 3G and Beyond", 2nd Edition, McGraw Hill Education (India) Pvt. Ltd, New Delhi, 2016.
2. Anwer Al-Dulaimi, Xianbin Wang and Chih-Lin I, "5G Networks: Fundamental Requirements, Enabling Technologies, and Operations Management", Wiley-IEEE Press, London, 2018.
3. Martin Sauter, "3G, 4G and Beyond: Bringing Networks, Devices and the Wen Together", 2nd Edition, John Wiley & Sons, United Kingdom, 2013.
4. Kurose and Ross, "Computer Networking - a top-down approach", 6th Edition, Pearson Education, 2017.
5. L. L. Peterson and B. S. Davie, "Computer Networks - a systems approach", 5th Edition, Morgan Kaufman, 2011.

CO-PO & PSO MAPPING

CO	PO												PSO		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
1	2	3	2	2		1			2	2			1	2	2
2	2	3	3	2		1			2	2	1		1	3	1
3	3	3	3	3	2	1			2	2	1		1	1	1
4	3	2	3	2		1			2	2			1	1	1
5	3	1	2	1	2	1			2	2	1	3		1	3
AVg.	2.6	2.4	2.6	2	0.8	1			2	2	0.6	0.6	0.8	1.6	1.6

1-low, 2-medium,3-high

23AD414	FOUNDATION IN DATA SCIENCE	L	T	P	C
		3	0	0	3

MODULE I FUNDAMENTALS IN DATA SCIENCE

9

Data science in real world - Need for data science – benefits and uses – facets of data – data science process – setting the research goal – retrieving data – cleansing, integrating, and transforming data – exploratory data analysis – build the models – presenting and building applications.

MODULE II DESCRIPTIVE ANALYTICS

9

Frequency distributions –Interpreting distributions – Graphs for Quantitative Data- describing variability – Variability for qualitative and ranked data - Normal distributions and Z scores – Correlation – Scatter plots – Regression – Interpretation of r^2 – Multiple regression – Regression toward the mean.

MODULE III INFERENCE STATISTICS

9

Populations – samples – random sampling – Sampling distribution- Hypothesis testing– Estimation – point estimate.

MODULE IV VARIANCE ANALYSIS

9

T-test for one sample – sampling distribution of t – t-test procedure – t-test for two independent samples – p-value – statistical significance – t-test for two related samples–Introduction to chi-square tests.

MODULE V PREDICTIVE ANALYTICS

9

Linear least squares –Regression using StatsModels – Multiple regression –Logistic regression –Time series analysis –Introduction to survival analysis.

COURSE COUTCOMES

At the end of the course, students will be able to

CO1: Understand the data science Lifecycle and Pre-processing

CO2: Describe the descriptive data analytics

CO3: Apply the concepts of inferential data analytics and variance

CO4: Evaluate formulas, create charts and use tools for data analytics and visualization.

CO5: Design predictive models from data

TOTAL : 45 PERIODS

TEXT BOOKS

1. David Cielien, Arno D. B. Meysman, and Mohamed Ali, "Introducing Data Science", Manning Publications, 2016
2. I.A.Dhote, " Fundamentals of Data Science and Anytics", Technical Publication, 2022

3. REFERENCE BOOKS

1. Foreman, J.W, "Data Smart: Using Data Science to Transform Information into Insight", John Wiley & Sons, 2013.
2. O'Neil, C. & Schutt, "R. Doing Data Science", O'Reilly Media Inc, 2014.
3. Emc Expert team, "Data Science and Big Data Analytics: Discovering, Analyzing, Visualizing and Presenting Data", EMC Education Services, John Wiley & Sons, 2015.
4. Sanjeev J. Wagh, Manisha S. Bhende, Anuradha D. Thakare, "Fundamentals of Data Science", CRC Press, 2022.

CO-PO & PSO MAPPING

CO	PO												PSO		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
1	2	2			2	1	1	1	1	2	1	2	3		3
2	1	1	1		2	2	1			3	3	3	2	3	
3	2	3	2	3	3	1	1			2	1	2		2	
4	2	2	3	3	3	2	1	1	1	3			1		3
5	1		3	3	3	3			2	2		3		2	
Avg.	1.6	2	2.2	3	2.6	1.8	1	1	1.3	2.4	1.6	2.5	2	2.3	3

1-low, 2-medium,3-high

23AD421	MACHINE LEARNING LABORATORY	L	T	P	C
		0	0	4	2

LIST OF EXPERIMENTS BASED ON THE FOLLOWING CONCEPTS

1. Implementation of FIND S Algorithm to find the most specific hypothesis based on a given set of training data samples.
2. Implementation of Candidate-Elimination algorithm
3. Linear Regression, Logistic Regression algorithmic implementation
4. Support Vector Machines implementation
5. Working philosophy of the decision tree based ID3 algorithm
6. Building Artificial Neural Network – Backpropagation algorithm
7. Experiments based on Naïve Bayesian Classifiers
8. Clustering datasets – EM, k-Means and k-Nearest neighbour algorithms
9. Fitting Data points – Locally weighted Regression algorithm

COURSE COUTCOMES

At the end of the course, students will be able to

CO1: Demonstrate various supervised learning paradigms for real-time data sets.

CO2: To possess and insight in the working philosophies of ML approaches.

CO3: Be able to design and implement various machine learning algorithms in a range of real-world applications.

CO4: Experiment with large number of datasets for unsupervised algorithms

CO5: To experiment and evaluate machine learning algorithms for different problem domain

TOTAL: 60 PERIODS

CO-PO & PSO MAPPING

CO	PO												PSO		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
1	3	3	3	3	3				2	2	2	3	3	3	3
2	3	3	3	3	3				2	2	2	3	3	3	3
3	3	3	3	3	3				2	2	2	3	3	3	3
4	3	3	3	3	3				2	2	2	3	3	3	3
5	3	3	3	3	3				2	2	2	3	3	3	3
AVg.	3	3	3	3	3				2	2	2	3	3	3	3

1-low, 2-medium,3-high

23AD422	WEB DEVELOPMENT LABORATORY	L	T	P	C
		0	0	4	2

LIST OF EXPERIMENTS BASED ON THE FOLLOWING CONCEPTS

1. Create a static web page with HTML5 tags.
2. Create responsive web page with all types of Cascading style sheets
3. Client Side Scripts using Java Script.
4. Installation of Apache Tomcat web server.
5. Write programs in Java using Servlets:
6. Write programs in Java using JSP
7. Programs Using Xml – Schema – Xslt/Xsl.
8. Programs using DOM and SAX parsers.
9. Write a program using NODE.js
10. Program using REACT framework

COURSE OUTCOMES

At the end of the course, students will be able to

CO1: Design simple web pages using markup languages like HTML and XHTML.

CO2: Create dynamic web pages using DHTML and java script that is easy to navigate and use

CO3: Experiment with Java using JSP, Servlets, XML/DOM/SAX parsers

TOTAL: 60 PERIODS

CO-PO & PSO MAPPING

CO	PO												PSO		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
1	3	3	3	2	3	1			2	2	3	3	2	3	3
2	3	3	3	3	3	1			2	2	3	3	2	3	3
3	3	3	3	3	3	1			2	2	3	3	2	3	3
AVg.	3	3	3	2.6	3	1			2	2	3	3	2	3	3

1-low, 2-medium,3-high

23AD423	STATISTICAL PROGRAMMING FOR DATA SCIENCE	L	T	P	C
		0	0	2	1

LIST OF EXPERIMENTS BASED ON THE FOLLOWING CONCEPTS

1. Basic Programming to understand the concepts of R programming
2. Arithmetic Operators
3. Logical Operations
4. Using Functions
5. Creating Variables, Numeric, Character and Logical Data
6. Vectors, Lists, Matrices, Arrays, Factors
7. Data Frames
8. Iteration and conditional statements
9. Use R functions and packages to extend the R programming environment.
10. Data analysis and Visualization: Loading and processing data, visually inspecting and analyzing using statistical estimates.

COURSE COUTCOMES

At the end of the course, students will be able to

CO1: Understand the basic syntax of R programming through practical exercises

CO2: Develop simple R programs using data types, iteration, control structures, functions, and Boolean operators

CO3: Apply exploratory data analysis using R programming and Visualize data attributes to identify hidden patterns from data

TOTAL: 30 PERIODS

CO-PO & PSO MAPPING

CO	PO												PSO		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
1	2	1	1	1	3	1	1	1	1	2	1	1	1	2	3
2	1	2	2	1	3	1	1	1	2	3	1	1	2	3	2
3	1	2	3	2	3	1	1	1	2	3	1	1	1	3	3
4	1	1	3	1	3	1	1	1	2	3	2	1	1	3	3
5	1	1	2	1	3	1	1	1	2	3	1	1	1	2	3
AVg.	1.2	1.4	2.2	1.2	3	1	1	1	1.8	2.8	1.2	1	1.2	2.6	2.8

1-low, 2-medium, 3-high